

Approval-Based Group-Fair Committee Selection: Proportional and Max-Min Objectives

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Abstract

In many decision-making settings, multiple evaluators express preferences over a common set of candidates and the final outcome is a selected committee or top- k subset. Classical approval-based rules simply choose the k candidates with highest approval scores, ignoring how representation is distributed across protected groups such as gender, race, or department. At the same time, institutions increasingly require group-fairness guarantees, not only in terms of minimum representation, but also in terms of how well each group is served by the chosen committee.

This paper introduces an approval-based framework for *group-fair committee selection* that combines ideas from proportional representation and max-min fairness. We work directly with approval ballots and protected groups, without using pairwise disagreement or rank-aggregation machinery. First, we adopt a Proportional Approval Voting (PAV) utility that balances individual representation and discourages domination by voters who approve many candidates. On top of this, we introduce group-level notions of representation and satisfaction, and study two complementary fairness paradigms: (i) proportional group representation via soft penalties for underrepresented groups, and (ii) max-min group satisfaction, which maximizes the welfare of the worst-off group. We formulate optimization problems that combine PAV utility with these group objectives, propose exact integer-programming formulations and practical greedy and local-search algorithms, and discuss structural and complexity properties. Our approach operates natively on approval ballots, focuses on committees rather than full rankings, and departs fundamentally from existing pairwise and parity-based ranking fairness formulations.

1 Introduction

Top- k selection is ubiquitous in hiring, admissions, recommendation, and many other domains where a fixed-size committee or shortlist of candidates must be chosen. Approval ballots—where each evaluator indicates a subset of acceptable candidates—offer a natural and widely used feedback model. Standard approval-based rules select the k candidates with highest approval scores, or apply proportional rules such as Proportional Approval Voting (PAV), but do not explicitly account for how representation and satisfaction are distributed across protected groups.

In parallel, there is a growing requirement to ensure that outcomes are *group-fair*: candidates and the groups they belong to should not be systematically disadvantaged due to protected attributes such as gender, race, or department. Moreover, fairness constraints may involve both hard representation requirements (e.g., at least a prescribed number of candidates from each group) and softer notions of how well each group is served in terms of total approval satisfaction.

Existing work on group-fairness in rankings or top- k selection often relies on full ranking ballots together with pairwise disagreement or parity-style exposure constraints. In contrast, we focus on the approval setting and on *committees* as the primary outcome:

- **Input:** approval ballots, possibly many approvals per voter;
- **Output:** a size- k committee (top- k subset) of candidates;
- **Objective:** maximize approval-based utility while ensuring fair treatment of multiple protected groups.

Contributions. Our main contributions are:

- We formulate a PAV-based approval utility for committees and extend it with group-level representation and satisfaction measures for multiple protected attributes and their intersections.
- We define two fairness paradigms: (i) proportional group representation via penalties for underrepresentation, and (ii) max-min group satisfaction, which maximizes the utility of the worst-off group.
- We propose hybrid optimization objectives that combine PAV utility with these group-fairness notions, and formalize the corresponding committee selection problems.
- We present integer linear programming (ILP) formulations and scalable greedy and local-search algorithms for these problems, and discuss their complexity and structural properties.

Our framework operates natively on approval ballots and works directly with committees, avoiding pairwise comparison machinery and ranking-first formulations.

2 Preliminaries

2.1 Candidates, voters, and approval ballots

Let $C = \{c_1, \dots, c_n\}$ be a set of n candidates and $V = \{v_1, \dots, v_m\}$ a set of m voters. Each voter v_i submits an *approval ballot* $A_i \subseteq C$, where $c \in A_i$ means that v_i approves candidate c . We denote by $\mathcal{A} = (A_1, \dots, A_m)$ the approval profile.

The *approval score* of a candidate c is

$$\text{score}(c) = \sum_{i=1}^m \mathbf{1}[c \in A_i].$$

A *committee* (or top- k subset) is a set $S \subseteq C$ with $|S| = k$ for some fixed committee size k . Our goal is to choose a committee S based on $\mathcal{A}$ that balances utility and group fairness.

2.2 Protected attributes and groups

Each candidate $c \in C$ is characterized by a fixed vector of protected attributes $(p_1(c), \dots, p_q(c))$. For each attribute p_k we denote its domain by $\text{dom}(p_k) = \{v_k^1, \dots, v_k^{r_k}\}$.

Definition 1 (Attribute group). For an attribute p_k and value $v \in \text{dom}(p_k)$, the corresponding attribute group is

$$G_{k,v} = \{c \in C : p_k(c) = v\}.$$

The Cartesian product of all attributes yields intersectional groups.

Definition 2 (Intersectional group). Let $\vec{v} = (v_1, \dots, v_q)$ with $v_k \in \text{dom}(p_k)$ for all k . The intersectional group associated with $\vec{v}$ is

$$G_{\vec{v}}^{\text{int}} = \{c \in C : p_k(c) = v_k \text{ for all } k \in \{1, \dots, q\}\}.$$

Depending on the application, fairness may be required at the level of attribute groups $\{G_{k,v}\}$, intersectional groups $\{G_{\vec{v}}^{\text{int}}\}$, or both.

3 Committee Utility from Approval Ballots

We now formalize how desirable a committee is from the viewpoint of the voters, ignoring group information.

3.1 Simple approval utility

A natural baseline is the total number of approvals in the committee:

$$U_{\text{sum}}(S) = \sum_{i=1}^m |A_i \cap S| = \sum_{c \in S} \text{score}(c).$$

This objective simply selects the k highest-scoring candidates.

3.2 Proportional Approval Voting (PAV) utility

To avoid over-rewarding voters who approve many candidates and to promote proportional representation, we adopt the Proportional Approval Voting (PAV) utility.

For a voter v_i and committee S , let $t_i(S) = |A_i \cap S|$ be the number of approved candidates in S . The PAV utility of S is

$$U_{\text{PAV}}(S) = \sum_{i=1}^m \sum_{j=1}^{t_i(S)} \frac{1}{j}.$$

The harmonic weights $\frac{1}{j}$ ensure diminishing returns for additional approved candidates per voter. Maximizing $U_{\text{PAV}}(S)$ over all $|S| = k$ is known to yield committees with desirable proportionality properties, but the optimization problem is NP-hard in general.

In what follows, $U(S)$ will denote either $U_{\text{PAV}}(S)$ or another approval-based utility function, depending on context.

4 Group Fairness from Approvals

We now introduce group-level fairness notions, defined directly on committees.

4.1 Group representation and shortfall

For a group $G \subseteq C$ and committee S , the number of seats allocated to G is $|S \cap G|$. Suppose that for each group G we are given a minimum desired representation $r_G \geq 0$, with the understanding that

$$\sum_G r_G \leq k$$

over any collection of groups considered simultaneously (e.g., all attribute groups for a single attribute, or a small set of intersectional groups).

Definition 3 (Representation shortfall). For group G and committee S , the representation shortfall of G is

$$\text{short}(G, S) = \max\{0, r_G - |S \cap G|\}.$$

The shortfall is zero if the group meets or exceeds its target, and positive otherwise.

4.2 Group satisfaction

Representation alone does not capture how well a group is served in terms of approvals. We therefore define group satisfaction as approval utility restricted to that group.

Definition 4 (Group satisfaction). For group G and committee S , the (raw) satisfaction of G is

$$U_G(S) = \sum_{i=1}^m |A_i \cap S \cap G|.$$

In words, $U_G(S)$ counts approvals in S that are assigned to candidates from G . We also consider a normalized version that accounts for group size or approval potential.

Let $U_G^{\max}$ be an upper bound on achievable satisfaction for group G (e.g., the maximum over all $|S| = k$ committees, or simply $\sum_i |A_i \cap G|$). We define normalized satisfaction as

$$\bar{U}_G(S) = \begin{cases} \frac{U_G(S)}{U_G^{\max}} & \text{if } U_G^{\max} > 0, \\ 0 & \text{if } U_G^{\max} = 0. \end{cases}$$

4.3 Proportional and max-min fairness objectives

We now introduce two key fairness paradigms.

Proportional representation via penalties. We define a global representation penalty as

$$\text{Pen}_{\text{rep}}(S) = \sum_{G \in \mathcal{G}} \text{short}(G, S),$$

where $\mathcal{G}$ is the collection of groups we enforce proportionality over (e.g., all attribute groups or a chosen set of intersectional groups). We then consider objectives of the form

$$\max_{S:|S|=k} U(S) - \lambda \cdot \text{Pen}_{\text{rep}}(S),$$

where $\lambda \geq 0$ is a trade-off parameter.

Max-min group satisfaction. To ensure that no group is left far behind in terms of satisfaction, we define:

$$\text{MM}(S) = \min_{G \in \mathcal{G}} \bar{U}_G(S).$$

The max-min objective is then

$$\max_{S:|S|=k} \text{MM}(S),$$

possibly combined with overall utility as a secondary criterion.

5 Optimization Problems

We now formalize the committee selection problems induced by these objectives.

Problem 1 (Proportional Approval-Fair Committee (PAF-Comm)). Given candidates C , voters V , approval profile $\mathcal{A}$, group set $\mathcal{G}$ with representation targets r_G for each $G \in \mathcal{G}$, committee size k , utility function $U(S)$ (e.g., PAV), and trade-off parameter $\lambda \geq 0$, find a committee $S \subseteq C$ with $|S| = k$ that maximizes

$$F_{\text{PAF}}(S) = U(S) - \lambda \cdot \text{Pen}_{\text{rep}}(S).$$

Problem 2 (Max-Min Group Approval Committee (MM-Comm)). Given the same inputs but without representation targets, find a committee S with $|S| = k$ that maximizes

$$F_{\text{MM}}(S) = \min_{G \in \mathcal{G}} \bar{U}_G(S).$$

In practice, it is attractive to combine these two fairness views.

Problem 3 (Hybrid Proportional + Max-Min Fair Committee). Given parameters $\lambda, \mu \geq 0$, find a committee S with $|S| = k$ that maximizes

$$F_{\text{hyb}}(S) = U(S) - \lambda \cdot \text{Pen}_{\text{rep}}(S) + \mu \cdot \text{MM}(S).$$

Alternatively, one may lexicographically maximize $F_{\text{MM}}(S)$ and then break ties by $U(S)$.

These problems generalize standard PAV (by setting $\lambda = \mu = 0$) and pure max-min group satisfaction (by setting $\lambda = 0$, prioritizing MM), while providing flexibility to tune trade-offs.

6 Algorithms

We now propose algorithmic approaches for Problems 1–3.

6.1 Integer programming formulation

We first give an integer linear programming formulation for PAF-Comm with PAV utility; the other variants can be obtained by adjusting the objective and constraints.

Let binary variables $x_c \in \{0, 1\}$ indicate committee membership: $x_c = 1$ if and only if $c \in S$. We enforce the committee size by

$$\sum_{c \in C} x_c = k.$$

To linearize PAV utility, we use auxiliary variables. For each voter i and integer $j \in \{1, \dots, k\}$, introduce a binary variable $z_{i,j} \in \{0, 1\}$ that indicates whether v_i receives at least j approved candidates in the committee. We then enforce:

$$\sum_{c \in A_i} x_c \geq \sum_{j=1}^k z_{i,j} \quad \text{for all } i.$$

The PAV utility is

$$U_{\text{PAV}}(S) = \sum_{i=1}^m \sum_{j=1}^k \frac{1}{j} z_{i,j}.$$

For group representation penalties, we define integer slack variables $s_G \geq 0$ for each group $G \in \mathcal{G}$, with

$$\sum_{c \in G} x_c + s_G \geq r_G,$$

so that $s_G \geq \text{short}(G, S)$ and we may set $s_G = \text{short}(G, S)$ in an optimal solution.

The PAF-Comm objective can then be written as

$$\max \quad \sum_{i=1}^m \sum_{j=1}^k \frac{1}{j} z_{i,j} - \lambda \sum_{G \in \mathcal{G}} s_G.$$

To incorporate max-min satisfaction in the hybrid problem, we introduce an additional continuous variable τ and constrain

$$\sum_{i=1}^m |A_i \cap G|^{-1} \sum_{c \in G} x_c \mathbf{1}[c \in A_i] \geq \tau \cdot U_G^{\max}$$

for each $G \in \mathcal{G}$, appropriately linearized. We then add $\mu\tau$ to the objective as in Problem 3.

The resulting ILP exactly encodes PAF-Comm and its hybrid variants, but is computationally challenging for large instances.

6.2 Greedy construction algorithms

For larger instances, we propose greedy algorithms that iteratively build a committee by adding one candidate at a time.

At step t , we have a partial committee S_{t-1} of size $t-1$. Let $P_t = C \setminus S_{t-1}$ be the remaining candidates. For each $c \in P_t$, we consider the marginal contributions:

$$\Delta U(c \mid S_{t-1}) = U(S_{t-1} \cup \{c\}) - U(S_{t-1}),$$

$$\Delta \text{Pen}_{\text{rep}}(c \mid S_{t-1}) = \text{Pen}_{\text{rep}}(S_{t-1} \cup \{c\}) - \text{Pen}_{\text{rep}}(S_{t-1}),$$

$$\Delta \text{MM}(c \mid S_{t-1}) = \text{MM}(S_{t-1} \cup \{c\}) - \text{MM}(S_{t-1}),$$

where MM may be approximated cheaply.

For PAF-Comm we choose

$$\text{score}_t(c) = \Delta U(c \mid S_{t-1}) - \lambda \cdot \Delta \text{Pen}_{\text{rep}}(c \mid S_{t-1}),$$

and pick $c_t = \arg \max_{c \in P_t} \text{score}_t(c)$.

For the hybrid objective we can additionally weigh the change in worst-group satisfaction:

$$\text{score}_t(c) = \Delta U(c \mid S_{t-1}) - \lambda \cdot \Delta \text{Pen}_{\text{rep}}(c \mid S_{t-1}) + \mu \cdot \Delta \text{MM}(c \mid S_{t-1}).$$

This greedy construction generalizes classical approval-based committee rules by discounting candidates whose inclusion would worsen representation or leave the worst-off group under-served.

6.3 Local search refinement

Starting from any initial committee S (e.g., produced by greedy or by the pure PAV rule), we can further improve the objective using local search.

At each iteration, we consider all single swaps $S' = S \setminus \{c\} \cup \{d\}$ with $c \in S$, $d \notin S$; if any swap strictly improves the chosen objective (PAF or hybrid), we move to the best-improving neighbor. We terminate when no improving swap exists.

This simple local search is guaranteed to terminate in a finite number of steps. In practice it converges quickly and often yields a significant improvement over pure greedy constructions, especially when fairness penalties are strong.

7 Discussion and Extensions

Our PAV- and max-min-based committee selection framework admits several extensions.

Multiple attributes and intersections. The group set $\mathcal{G}$ can contain attribute groups for several protected attributes and/or intersectional groups. Representation targets r_G and normalization constants $U_G^{\max}$ can be specified separately for each group, allowing flexible policy definitions.

Hard vs. soft constraints. Instead of penalty terms, some groups may require hard lower bounds $|S \cap G| \geq r_G$ that must be satisfied exactly. These can be enforced directly in the ILP, while penalties are used for softer desiderata.

Margin-of-fairness. Building on the notion of approval substitutions, one can define the margin-of-fairness of a committee as the minimum number of approval changes required to violate a given fairness criterion. Analyzing this margin for PAF-Comm and MM-Comm is an interesting direction for future work.

Alternative utilities. Beyond PAV, other concave functions of $t_i(S)$ could be used, or combinations of PAV with domain-specific utility measures (e.g., quality scores or distances).

8 Conclusion

We have introduced a framework for group-fair committee selection from approval ballots, based on combining Proportional Approval Voting utility with group-level proportional and max-min fairness objectives. Our formulation operates directly on committees and approval data, departing from pairwise and ranking-based fairness models. We presented both exact ILP formulations and scalable greedy and local-search algorithms for optimizing these objectives.

Future work includes sharper approximation guarantees, detailed complexity analysis for specific parameter regimes, integration with margin-of-fairness concepts for robustness, and empirical evaluation of PAF-Comm and hybrid methods on real-world approval datasets.